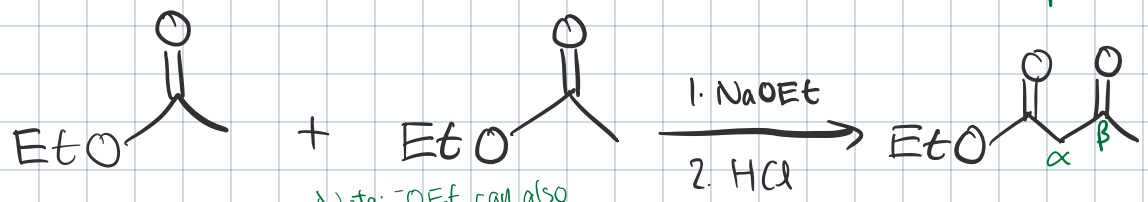
"β-keto ester"

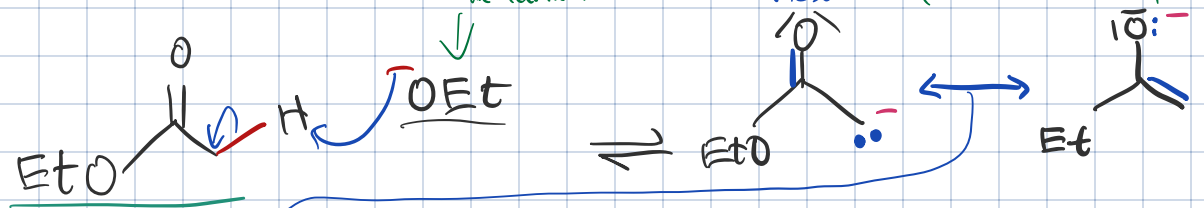


Note: ⁻OEt can also act as a NuPh through trans-esterification, but this won't change anything b/c the NuPh and original ester substituent are identical.

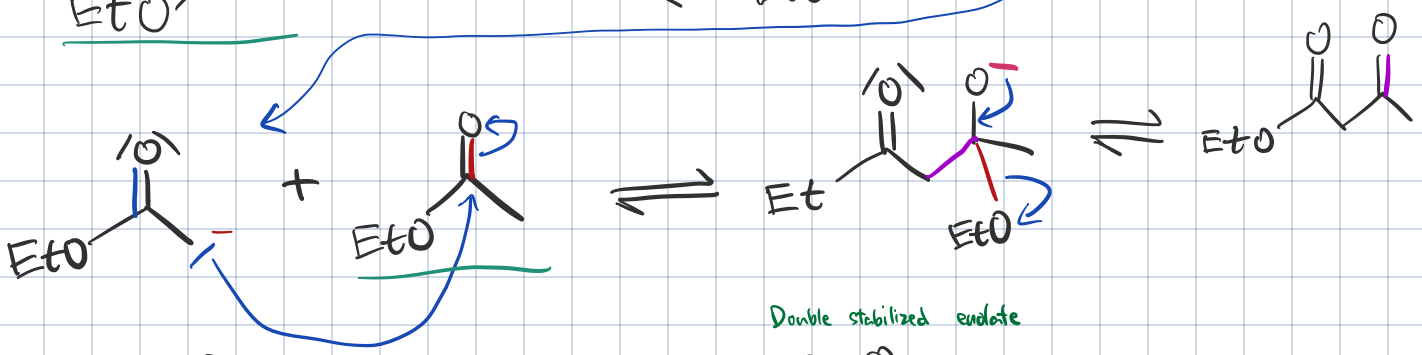
Mech.

Enolate/Carbanion Resonance (see Aldol rxn if needed)

1.

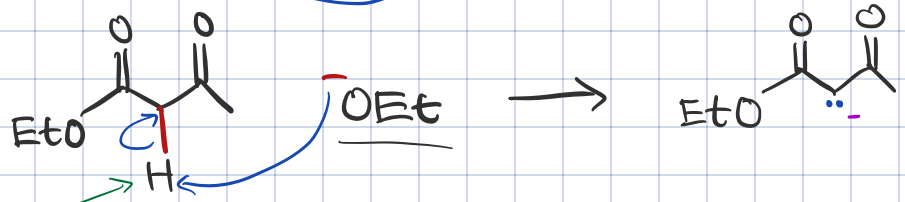


2.



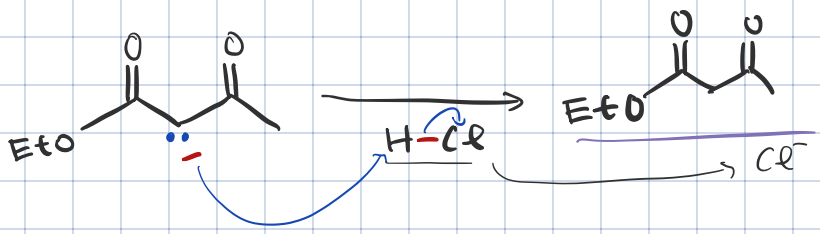
Double stabilized enolate

3.



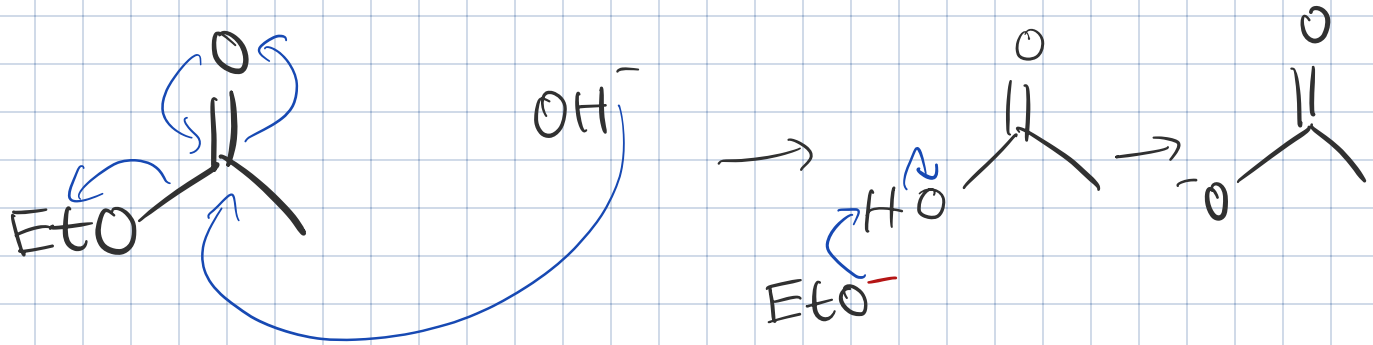
pretty acidic

4. workup

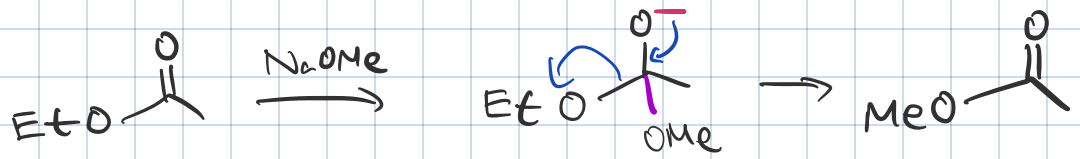


Selection of Base is important
If hydroxide is used instead of ethoxides ester hydrolysis may occur !!!

OH^- as base → hydrolysis

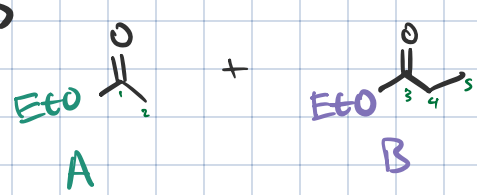


Methoxide as base → trans-esterification



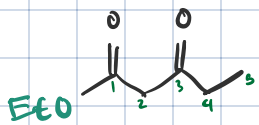
End result is a mixture:

Cross

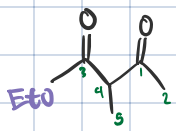


→ AA, AB, BA, BB — very similar to cross-aldol

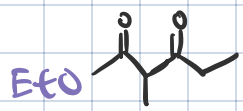
AB:



BA:



BB:



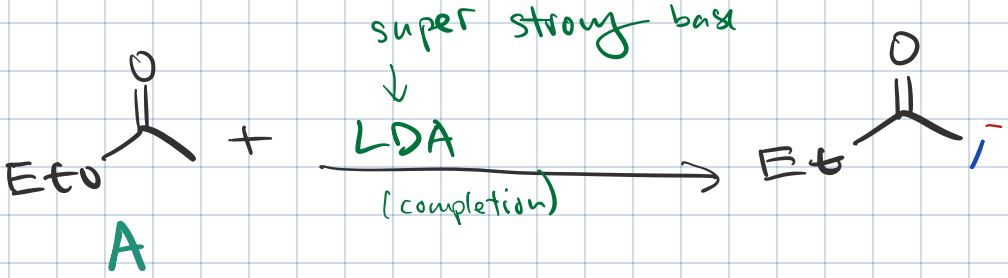
Selectivity

Ethoxide only partially deprotonates, LDA proceeds to completion

If we only want A → B for example, use LDA

↓

1:1 eq w/ A



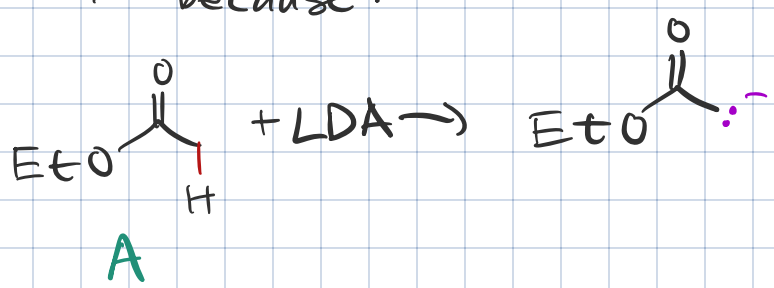
super strong base

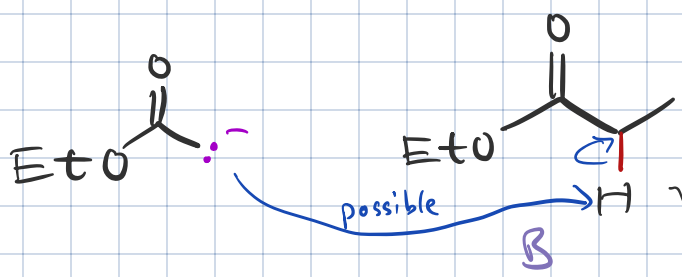
Prevents A from reacting w/ itself b/c they are all deprotonated → neither are electrophiles

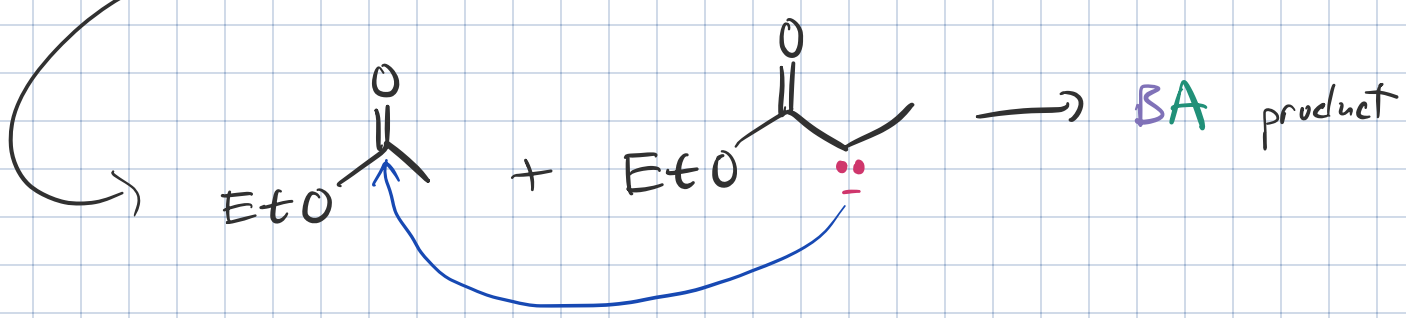
Thus,

$$A \xrightarrow[\text{2. B in 1:1 ratio w/ A}]{\text{1. LDA in 1:1 ratio w/ A}} AB \text{ product}$$

Not 100 % because:

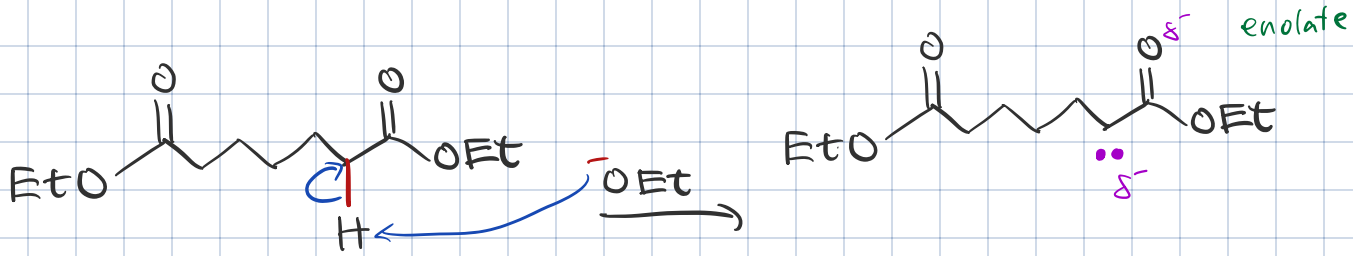


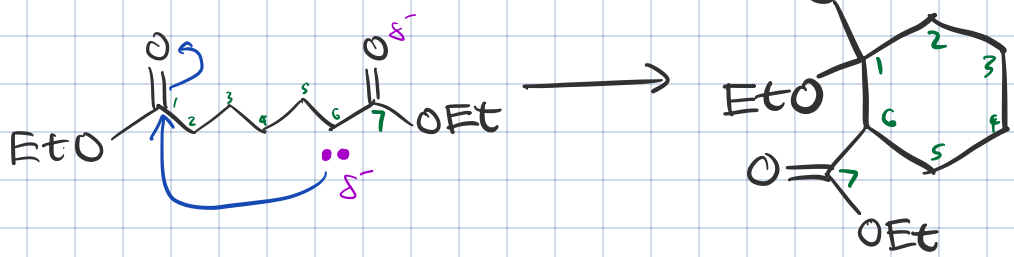


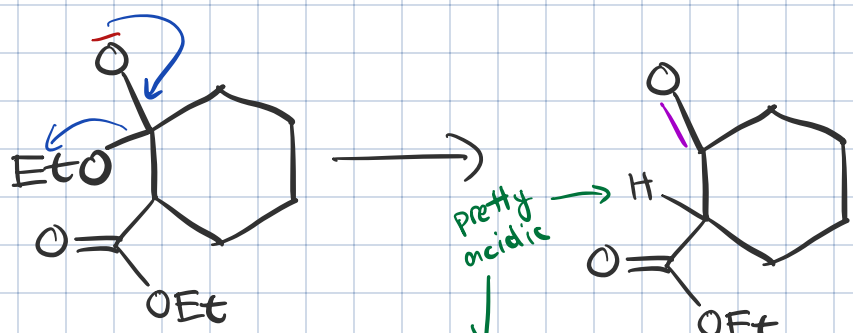


The LDA selective method will improve the yield, but you may still get a mixture of the other products

Dieckmann → An intramolecular Claisen Condensation

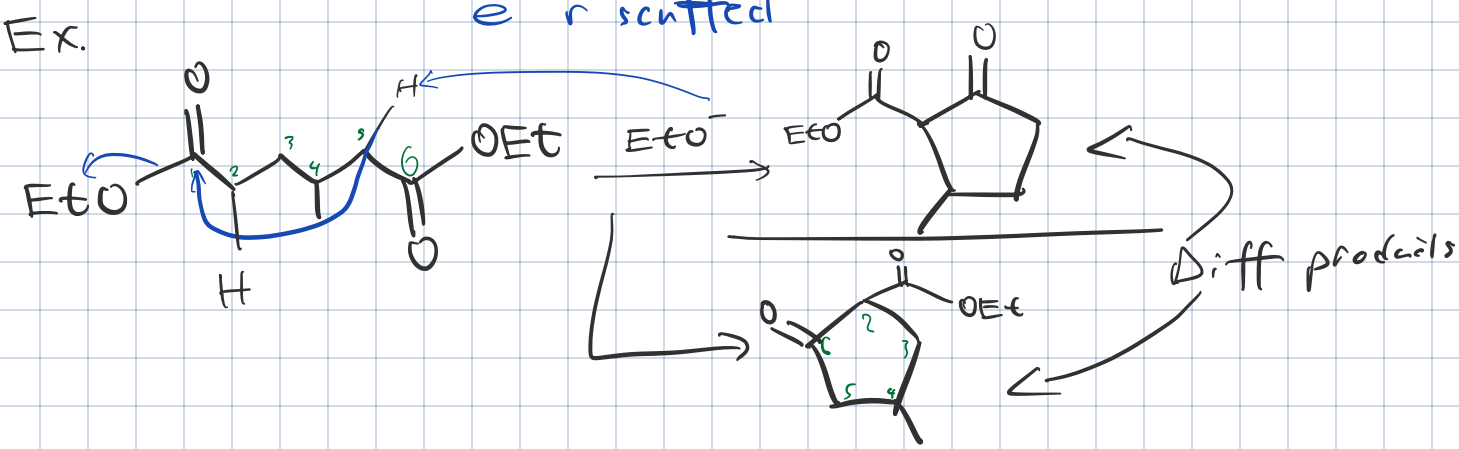






(can be further hydrolyzed and decarboxylated into cyclohexanone)

add HCl in the workup to protonate (EtO^- in solution as base)

Ex.

e^- r scuffed